

Student Name:	Amit Kumar	Roll No:	103	BARCODE BAR005
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Answer to Q1(i)

A computer takes input processes it and gives output. Main parts are input unit like keyboard, CPU which process data, memory to store, output unit like monitor. All connected together to form computer system.

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Answer to Q1(ii)

RAM is primary memory. Fast but temporary. Loses data on power off. Hard disk stores data permanently. Slower than RAM. RAM costs more per GB. RAM: 4GB, 8GB sticks. HDD: 500GB, 1TB drives.

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Answer to Q1(iii)

OS controls computer resources. It does process management, memory management and file management. Also provides user interface. Windows and Linux are examples of OS. OS is loaded first when computer starts.

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BAR005**Answer to Q2(i)**

Von Neumann architecture: Input → CPU → Output. CPU has CU and ALU. Memory connected to CPU. Bus system transfers data between parts. All instructions stored in memory and fetched one by one.

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Answer to Q3(i)

255 binary = 11111111. Octal = 377. Hex = FF. Each group of 3 binary bits = 1 octal digit. Each group of 4 binary bits = 1 hex digit.

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Answer to Q4(i)

Types of computers based on size: Super, Mainframe, Mini, Micro, Embedded. Super computer used for weather, science. Mainframe for banks. Personal computers are microcomputers. Embedded in appliances.

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Answer to Q7(i)

SRAM: uses flip flop circuits, fast, no refresh needed, costly, used in cache. DRAM: uses capacitors, needs refresh, slow, cheap, large capacity. Main memory uses DRAM. Cache uses SRAM.

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